

Theory

R provides file I/O functions to persist and retrieve data, and loop control structures to iterate over that data programmatically. Combining these two capabilities allows processing of real-world datasets stored in files.

File Operations:

- `write.table()` / `read.table()` — export/import tab-separated .txt files.
- `write.csv()` / `read.csv()` — export/import comma-separated .csv files.
- `list.files()` — verify files exist in the working directory.

Loop Control Structures:

- for loop — iterates over each row/element for a fixed number of times. Used for row-wise operations on data frames.
- while loop — repeats as long as a condition holds TRUE. Useful for conditional traversal or threshold-based stopping.
- repeat loop — runs indefinitely until `break` is called. Used when exit condition is evaluated inside the body.
- `next` — skips the current iteration and proceeds to the next.
- `break` — exits the loop immediately, regardless of remaining iterations.

Part I — Create Student Dataset and Export to TXT & CSV

Code for Part i — Create Data Frame and Export to students.txt and students.csv:

```
RStudio
File Edit Code View Plots SessionBuild DebugProfile Tools Help
4a_create_files.R
Source on Save Run Source
1 # Practical 4 - Part i: Create Student Dataset and Export to TXT & CSV
2
3 # Step 1: Set working directory
4 setwd("C:/Users/Student/R_Practicals")
5
6 # Step 2: Create student data frame
7 student_data <- data.frame(
8   RollNo = c(101, 102, 103, 104, 105, 106),
9   Name    = c("Aarav Sharma", "Priya Desai", "Rahul Patil",
10              "Sneha Joshi", "Amit Verma", "Neha Kulkarni"),
11   Age     = c(21, 22, 20, 23, 21, 22),
12   Marks   = c(88, 76, 92, 65, 81, 57),
13   Grade   = c("A", "B", "A", "C", "B", "D"),
14   City    = c("Nagpur", "Mumbai", "Pune",
15              "Nagpur", "Delhi", "Aurangabad")
16 )
17
18 # Step 3: Display the data frame
19 cat("--- Student Dataset ---\n")
20 print(student_data)
21
22 # Step 4: Export to TXT (tab-separated)
23 write.table(student_data, file = "students.txt", sep = "\t", row.names = FALSE, quote = FALSE)
24 cat("\nExported to students.txt\n")
25
26 # Step 5: Export to CSV
27 write.csv(student_data, file = "students.csv", row.names = FALSE)
28 cat("Exported to students.csv\n")
29
30 # Step 6: Verify files created
31 cat("\nFiles in directory:\n")
32 cat(list.files(), sep = "\n")
```

Output for Part i:

```
Console ~/
> cat("--- Student Dataset ---\n")
--- Student Dataset ---
  RollNo      Name Age Marks Grade      City
1    101  Aarav Sharma 21   88    A    Nagpur
2    102  Priya Desai  22   76    B    Mumbai
3    103  Rahul Patil  20   92    A     Pune
4    104  Sneha Joshi  23   65    C    Nagpur
5    105  Amit Verma  21   81    B     Delhi
6    106  Neha Kulkarni 22   57    D Aurangabad

> cat("\nExported to students.txt\n")

Exported to students.txt
> cat("Exported to students.csv\n")
Exported to students.csv

> cat("\nFiles in directory:\n")

Files in directory:
grade_A_students.csv
students.csv
students.txt
```

Part II — Loop Control Structures on Text File (students.txt)

Program 4b — for Loop on students.txt

Code for Program 4b — for Loop on TXT File:

```
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4b_for_loop_txt.R
Source on Save ▶ Run Source ▼

1 # Practical 4 - Part ii: for Loop on TXT File
2
3 # Step 1: Import TXT file
4 txt_data <- read.table("students.txt", header = TRUE, sep = "\t")
5 cat("--- Data from students.txt ---\n")
6 print(txt_data)
7
8 # Step 2: for loop - print each student's name and marks
9 cat("\n--- for loop: Student Report (TXT) ---\n")
10 for (i in 1:nrow(txt_data)) {
11   cat("Roll:", txt_data$RollNo[i],
12       " | Name:", txt_data$Name[i],
13       " | Marks:", txt_data$Marks[i],
14       " | Grade:", txt_data$Grade[i], "\n")
15 }
16
17 # Step 3: for loop - classify Pass/Fail
18 cat("\n--- for loop: Pass/Fail Classification ---\n")
19 for (i in 1:nrow(txt_data)) {
20   result <- ifelse(txt_data$Marks[i] >= 60, "Pass", "Fail")
21   cat(txt_data$Name[i], ":", result, "\n")
22 }
23
24 # Step 4: for loop - calculate total and average marks
25 cat("\n--- for loop: Total & Average Marks ---\n")
26 total <- 0
27 for (m in txt_data$Marks) {
28   total <- total + m
29 }
30 cat("Total Marks :", total, "\n")
31 cat("Average Marks:", total / nrow(txt_data), "\n")
```

Output for Program 4b:

```
Console ~/
> cat("--- Data from students.txt ---\n")
--- Data from students.txt ---
  RollNo      Name Age Marks Grade      City
1   101   Aarav Sharma  21   88    A   Nagpur
2   102   Priya Desai  22   76    B   Mumbai
3   103   Rahul Patil  20   92    A    Pune
4   104   Sneha Joshi  23   65    C   Nagpur
5   105   Amit Verma   21   81    B    Delhi
6   106   Neha Kulkarni 22   57    D Aurangabad

--- for loop: Student Report (TXT) ---
Roll: 101 | Name: Aarav Sharma | Marks: 88 | Grade: A
Roll: 102 | Name: Priya Desai | Marks: 76 | Grade: B
Roll: 103 | Name: Rahul Patil | Marks: 92 | Grade: A
Roll: 104 | Name: Sneha Joshi | Marks: 65 | Grade: C
Roll: 105 | Name: Amit Verma | Marks: 81 | Grade: B
Roll: 106 | Name: Neha Kulkarni | Marks: 57 | Grade: D

--- for loop: Pass/Fail Classification ---
Aarav Sharma : Pass
Priya Desai : Pass
Rahul Patil : Pass
Sneha Joshi : Pass
Amit Verma : Pass
Neha Kulkarni : Fail

--- for loop: Total & Average Marks ---
Total Marks : 459
Average Marks: 76.5
```

Program 4c — while Loop on students.txt

Code for Program 4c — while Loop on TXT File:

```
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4c_while_loop_txt.R
Source on Save ▶ Run Source ▼

1 # Practical 4 - Part ii: while Loop on TXT File
2
3 # Step 1: Import TXT file
4 txt_data <- read.table("students.txt", header = TRUE, sep = "\t")
5
6 # Step 2: while loop – print students one by one
7 cat("--- while loop: Student List (TXT) ---\n")
8 i <- 1
9 while (i <= nrow(txt_data)) {
10   cat("[", i, "]", txt_data$Name[i], "- City:", txt_data$City[i], "\n")
11   i <- i + 1
12 }
13
14 # Step 3: while loop – find first student with Marks >= 90
15 cat("\n--- while loop: First student with Marks >= 90 ---\n")
16 i <- 1
17 while (i <= nrow(txt_data)) {
18   if (txt_data$Marks[i] >= 90) {
19     cat("Topper:", txt_data$Name[i], "with Marks =", txt_data$Marks[i], "\n")
20     break
21   }
22   i <- i + 1
23 }
24
25 # Step 4: while loop – running total until total > 200
26 cat("\n--- while loop: Running Total (stop > 200) ---\n")
27 total <- 0; i <- 1
28 while (total <= 200 && i <= nrow(txt_data)) {
29   total <- total + txt_data$Marks[i]
30   cat("Added", txt_data$Name[i], "(", txt_data$Marks[i], ") | Total:", total, "\n")
31   i <- i + 1
32 }
33 cat("Loop stopped. Final Total =", total, "\n")
```

Output for Program 4c:

```
Console ~/
> cat("--- while loop: Student List (TXT) ---\n")
--- while loop: Student List (TXT) ---
[ 1 ] Aarav Sharma - City: Nagpur
[ 2 ] Priya Desai - City: Mumbai
[ 3 ] Rahul Patil - City: Pune
[ 4 ] Sneha Joshi - City: Nagpur
[ 5 ] Amit Verma - City: Delhi
[ 6 ] Neha Kulkarni - City: Aurangabad

--- while loop: First student with Marks >= 90 ---
Topper: Rahul Patil with Marks = 92

--- while loop: Running Total (stop > 200) ---
Added Aarav Sharma ( 88 ) | Total: 88
Added Priya Desai ( 76 ) | Total: 164
Added Rahul Patil ( 92 ) | Total: 256
Loop stopped. Final Total = 256
```

Program 4d — repeat Loop on students.txt

Code for Program 4d — repeat Loop on TXT File:

```
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4d_repeat_loop_txt.R
Source on Save ▶ Run Source ▼

1 # Practical 4 - Part ii: repeat Loop on TXT File
2
3 # Step 1: Import TXT file
4 txt_data <- read.table("students.txt", header = TRUE, sep = "\t")
5
6 # Step 2: repeat loop – print all student names
7 cat("--- repeat loop: All Student Names (TXT) ---\n")
8 i <- 1
9 repeat {
10   cat("Student", i, ":", txt_data$Name[i], "\n")
11   i <- i + 1
12   if (i > nrow(txt_data)) break
13 }
14
15 # Step 3: repeat loop – find students below average
16 cat("\n--- repeat loop: Below Average Students ---\n")
17 avg <- mean(txt_data$Marks)
18 cat("Class Average:", avg, "\n")
19 i <- 1
20 repeat {
21   if (txt_data$Marks[i] < avg)
22     cat("Below Avg:", txt_data$Name[i], "| Marks:", txt_data$Marks[i], "\n")
23   i <- i + 1
24   if (i > nrow(txt_data)) break
25 }
26
27 # Step 4: repeat with next – print only Grade A students
28 cat("\n--- repeat loop: Grade A Students ---\n")
29 i <- 1
30 repeat {
31   if (txt_data$Grade[i] != "A") { i <- i + 1; if (i > nrow(txt_data)) break; next }
32   cat(txt_data$Name[i], "- Marks:", txt_data$Marks[i], "\n")
33   i <- i + 1
34   if (i > nrow(txt_data)) break
35 }
```

Output for Program 4d:

```
Console ~/
> cat("--- repeat loop: All Student Names (TXT) ---\n")
--- repeat loop: All Student Names (TXT) ---
Student 1 : Aarav Sharma
Student 2 : Priya Desai
Student 3 : Rahul Patil
Student 4 : Sneha Joshi
Student 5 : Amit Verma
Student 6 : Neha Kulkarni

--- repeat loop: Below Average Students ---
Class Average: 76.5
Below Avg: Priya Desai | Marks: 76
Below Avg: Sneha Joshi | Marks: 65
Below Avg: Neha Kulkarni | Marks: 57

--- repeat loop: Grade A Students ---
Aarav Sharma - Marks: 88
Rahul Patil - Marks: 92
```

Part II — Loop Control Structures on CSV File (students.csv)

Program 4e — for Loop on students.csv

Code for Program 4e — for Loop on CSV File:

```
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4e_for_loop_csv.R
Source on Save ▶ Run Source ▼

1 # Practical 4 - Part ii: for Loop on CSV File
2
3 # Step 1: Import CSV file
4 csv_data <- read.csv("students.csv")
5 cat("--- Data from students.csv ---\n")
6 print(csv_data)
7
8 # Step 2: for loop - display roll number and city
9 cat("\n--- for loop: RollNo & City (CSV) ---\n")
10 for (i in 1:nrow(csv_data)) {
11   cat("RollNo:", csv_data$RollNo[i], " | City:", csv_data$City[i], "\n")
12 }
13
14 # Step 3: for loop - grade-wise count
15 cat("\n--- for loop: Grade-wise Count (CSV) ---\n")
16 grades <- c("A", "B", "C", "D")
17 for (g in grades) {
18   count <- sum(csv_data$Grade == g)
19   cat("Grade", g, ": ", count, "student(s)\n")
20 }
21
22 # Step 4: for loop - export filtered data (Marks > 70)
23 cat("\n--- for loop: Students with Marks > 70 ---\n")
24 filtered <- data.frame()
25 for (i in 1:nrow(csv_data)) {
26   if (csv_data$Marks[i] > 70) {
27     filtered <- rbind(filtered, csv_data[i, ])
28   }
29 }
30 print(filtered)
31 write.csv(filtered, "marks_above_70.csv", row.names = FALSE)
32 cat("Exported to marks_above_70.csv\n")
```

Output for Program 4e:

```
Console ~/
> cat("--- Data from students.csv ---\n")
--- Data from students.csv ---
  RollNo      Name Age Marks Grade      City
1    101  Aarav Sharma  21    88     A   Nagpur
2    102   Priya Desai  22    76     B   Mumbai
3    103   Rahul Patil  20    92     A     Pune
4    104   Sneha Joshi  23    65     C   Nagpur
5    105    Amit Verma  21    81     B     Delhi
6    106   Neha Kulkarni 22    57     D Aurangabad

--- for loop: RollNo & City (CSV) ---
RollNo: 101 | City: Nagpur
RollNo: 102 | City: Mumbai
RollNo: 103 | City: Pune
RollNo: 104 | City: Nagpur
RollNo: 105 | City: Delhi
RollNo: 106 | City: Aurangabad

--- for loop: Grade-wise Count (CSV) ---
Grade A :  2 student(s)
Grade B :  2 student(s)
Grade C :  1 student(s)
Grade D :  1 student(s)

--- for loop: Students with Marks > 70 ---
  RollNo      Name Age Marks Grade      City
1    101  Aarav Sharma  21    88     A   Nagpur
2    102   Priya Desai  22    76     B   Mumbai
3    103   Rahul Patil  20    92     A     Pune
5    105    Amit Verma  21    81     B     Delhi
Exported to marks_above_70.csv
```

Program 4f — while & repeat Loops on students.csv

Code for Program 4f — while and repeat Loops on CSV File:

```
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4f_while_repeat_csv.R
Source on Save ▶ Run Source ▼

1 # Practical 4 - Part ii: while & repeat Loops on CSV File
2
3 # Step 1: Import CSV file
4 csv_data <- read.csv("students.csv")
5
6 # Step 2: while loop — Fibonacci index traversal on CSV
7 cat("--- while loop: Age Verification (CSV) ---\n")
8 i <- 1
9 while (i <= nrow(csv_data)) {
10   status <- ifelse(csv_data$Age[i] >= 21, "Senior", "Junior")
11   cat(csv_data$Name[i], "(Age:", csv_data$Age[i], ") ->", status, "\n")
12   i <- i + 1
13 }
14
15 # Step 3: while loop — accumulate until top 3 marks summed
16 cat("\n--- while loop: Sum of Top 3 Marks ---\n")
17 sorted_marks <- sort(csv_data$Marks, decreasing = TRUE)
18 total <- 0; i <- 1
19 while (i <= 3) {
20   total <- total + sorted_marks[i]
21   cat("Rank", i, "Marks:", sorted_marks[i], "| Running Sum:", total, "\n")
22   i <- i + 1
23 }
24 cat("Sum of Top 3 Marks =", total, "\n")
25
26 # Step 4: repeat loop — display city-wise students from CSV
27 cat("\n--- repeat loop: Nagpur Students (CSV) ---\n")
28 i <- 1
29 repeat {
30   if (csv_data$City[i] == "Nagpur")
31     cat(csv_data$Name[i], "| Marks:", csv_data$Marks[i], "\n")
32   i <- i + 1
33   if (i > nrow(csv_data)) break
34 }
```

Output for Program 4f:

```
Console ~/
> cat("--- while loop: Age Verification (CSV) ---\n")
--- while loop: Age Verification (CSV) ---
Aarav Sharma (Age: 21 ) -> Senior
Priya Desai (Age: 22 ) -> Senior
Rahul Patil (Age: 20 ) -> Junior
Sneha Joshi (Age: 23 ) -> Senior
Amit Verma (Age: 21 ) -> Senior
Neha Kulkarni (Age: 22 ) -> Senior

--- while loop: Sum of Top 3 Marks ---
Rank 1 Marks: 92 | Running Sum: 92
Rank 2 Marks: 88 | Running Sum: 180
Rank 3 Marks: 81 | Running Sum: 261
Sum of Top 3 Marks = 261

--- repeat loop: Nagpur Students (CSV) ---
Aarav Sharma | Marks: 88
Sneha Joshi | Marks: 65
```

Conclusion

Thus, the following R programs for Practical 4 were successfully written, executed in RStudio, and outputs were verified:

- Part i: A 6-record student dataset (RollNo, Name, Age, Marks, Grade, City) was created as an R data frame and exported to students.txt using write.table() (tab-separated) and to students.csv using write.csv(). Both files were confirmed in the working directory using list.files().
- Program 4b (for loop on TXT): Imported students.txt using read.table(). Used for loop to print each student's roll number, name, marks, and grade; classify Pass/Fail based on marks ≥ 60 ; and compute total (459) and average (76.5) marks.
- Program 4c (while loop on TXT): Used while loop to traverse the dataset and print name-city pairs; find the first student with marks ≥ 90 (Rahul Patil — 92) using break; and accumulate a running total that stops once total exceeds 200.
- Program 4d (repeat loop on TXT): Used repeat loop to print all student names sequentially; identify below-average students (Priya, Sneha, Neha); and filter Grade A students using next to skip non-A grades.
- Program 4e (for loop on CSV): Imported students.csv using read.csv(). Used for loop to display RollNo-City pairs, count students per grade (A:2, B:2, C:1, D:1), filter students with Marks > 70 using rbind(), and export the filtered result to marks_above_70.csv.
- Program 4f (while & repeat on CSV): Used while loop for age-based classification (Senior/Junior) and to sum the top 3 marks ($92+88+81=261$). Used repeat loop to extract and display only Nagpur-based students from the CSV.

All loop control structures — for, while, and repeat — were successfully applied to both text and CSV file data in R, demonstrating their practical utility in real-world data processing tasks.